

Vishal Kumar Mathuri

RUST DEVELOPER | SOLANA / ANCHOR | CROSS-CHAIN & BLOCKCHAIN ENGINEERING

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SUMMARY

Blockchain Developer specializing in **Rust and the Solana ecosystem**, with hands-on experience building on-chain programs using Anchor, PDA-based account design, and cross-chain interoperability with LayerZero. Also experienced in Solidity and the Ethereum ecosystem, bringing a well-rounded, multi-chain engineering background across DeFi, NFTs, crowdfunding, and cross-chain messaging protocols.

Focused on **production-oriented Solana program design** — deterministic account derivation, secure authorization patterns, and end-to-end testing — with a growing portfolio of Anchor-based protocols spanning crowdfunding, cross-chain messaging, NFTs, and staking.

TECHNICAL SKILLS

Rust / Solana: Rust, Solana, Anchor, PDAs, SPL Token, Metaplex, Solana Web3.js, TypeScript, wallet integration

Cross-Chain: LayerZero V2, cross-chain messaging, relay architecture, message verification, replay protection

Solidity / EVM: Solidity, Ethereum, EVM, ERC-20, ERC-721, OpenZeppelin, Foundry, Hardhat

Security / Testing: unit testing, fuzz testing, invariant testing, account validation, authorization rules

Web3 / Backend: Ethers.js, React.js, Node.js, Express.js, REST APIs, MetaMask, Phantom Wallet

Tools: Git, GitHub, VS Code, Vercel, Render

PROFESSIONAL EXPERIENCE

Blockchain Developer Freelance	Jul 2025 - Present
- Developed and deployed 50+ Solidity smart contracts implementing ERC-20, ERC-721, staking, governance, and marketplace functionality across NFT, fundraising, identity, and access-control applications. - Built 10+ full-stack Web3 applications integrating smart contracts with React.js frontends, Node.js backend services, REST APIs, wallets, and EVM networks. - Applied OpenZeppelin security primitives and secure design patterns including reentrancy protection and role-based access control, reducing security risk across deployed contracts. - Worked across contract implementation, testing, deployment, frontend integration, debugging, and production issue resolution.	
Senior Content Developer Think & Learn Pvt Ltd	Jul 2019 - Mar 2023 Bengaluru, IN
- Managed scheduling operations for 200+ student batches across multiple grade levels with 100% operational accuracy and enhanced workflow efficiency. - Improved an automated concept-tagging workflow to 95% classification accuracy through systematic validation and quality checks, boosting system efficiency and reducing errors. - Crafted and executed structured Learning Journeys for academic programs utilizing digital learning platforms, resulting in a 96% success rate. - Designed and curated interactive educational content, mock tests, and learning modules for Byju's TV and BHELP programs, increasing user satisfaction and engagement. - Resolved student support and academic operations requests using Salesforce with 98% resolution accuracy within 24 hours, ensuring uninterrupted course delivery. - Collaborated with academic operations, HR, and support teams while maintaining strong cross-team communication and coordination.	

FLAGSHIP RUST & SOLANA PROJECTS

1. Solana Crowdfunding Protocol

Tech Stack: Rust • Solana • Anchor • PDAs • TypeScript • Wallet Adapter

Links: [GitHub](#) • [Live \(Vercel\)](#)

- Built a Solana crowdfunding protocol in Rust and Anchor supporting campaign creation, SOL contributions, funding-goal tracking, deadline enforcement, creator withdrawals, and contributor refunds.
- Designed **PDA-based campaign and protocol state**, using deterministic account derivation to associate campaign data and protocol-controlled accounts with campaign creators without relying on externally generated private keys.
- Implemented Anchor account constraints and authorization rules to validate campaign ownership, signer permissions, contribution state, withdrawal eligibility, and refund conditions at the program level.
- Delivered a TypeScript/Wallet Adapter integration layer for wallet-connected campaign interactions, with tests covering successful funding, failed campaigns, unauthorized withdrawals, deadline boundaries, and refund behavior.

2. LayerZero Cross-Chain Protocol — Solana/Rust Program (Ethereum ↔ Solana)

Tech Stack: Rust • Anchor • Solana • LayerZero V2 • Solana Web3.js

Links: [GitHub](#) • [Live Demo](#)

- Built the Solana-side program (Rust/Anchor) of a bidirectional **LayerZero V2 interoperability protocol** connecting Ethereum Sepolia and Solana Devnet, handling cross-chain application messaging and OFT (Omnichain Fungible Token) transfers.
- Implemented Solana transaction simulation and replay protection, along with a Phantom ALT (Address Lookup Table) compatibility workaround to handle Solana transaction size constraints.
- Built peer validation and message verification logic in Anchor, coordinating versioned transaction handling with the EVM-side Solidity contracts.
- Delivered end-to-end testing and deployment on Solana Devnet as part of a full cross-chain flow validated against Ethereum Sepolia.

3. Solana AMM / DEX Protocol

Tech Stack: Rust • Anchor • Solana • SPL Token • TypeScript

Status: In Development

- Building a Solana automated market maker (AMM) program in Rust and Anchor, supporting liquidity pool creation, token swaps, and constant-product pricing between SPL token pairs.
- Designing **PDA-based liquidity pool and vault accounts** to custody pooled tokens and track liquidity provider (LP) shares.
- Implementing swap execution logic with slippage protection, fee accrual, and proportional LP token minting/burning on deposits and withdrawals.
- Developing a TypeScript client and wallet integration layer for adding/removing liquidity and executing token swaps against the on-chain pool.

4. Solana Staking & Rewards Protocol

Tech Stack: Rust • Anchor • Solana • SPL Token • TypeScript

Status: In Development

- Building a Solana staking and rewards protocol in Rust and Anchor, enabling users to stake SPL tokens and earn time-based rewards from a program-controlled reward pool.
- Designing **PDA-based stake accounts and vault logic** to securely custody staked tokens and track individual stake positions.
- Implementing reward-rate calculations, unstaking cooldowns, and claim logic to ensure accurate and manipulation-resistant reward distribution.
- Developing a TypeScript client and wallet integration layer for staking, unstaking, and claiming rewards directly from a connected wallet.

INTERNSHIPS

Blockchain Developer
EtherAuthority

Mar 2026 - May 2026
Remote

- Built a Telegram Mini App featuring multiple in-app mini-games, including Coinflip, Find the King (a dice-based game), and Wheel, using the Telegram Web App platform.
- Designed and implemented game logic, betting flow, and result computation for each game mode.
- Collaborated with the team on frontend/backend integration and testing to ship engaging, low-latency gameplay experiences.

Blockchain Developer
Leafny Web Pack

May 2025 - Jul 2025
Remote

- Assisted in building Web3 dApps integrating smart contracts with frontend and backend systems.
- Worked with ERC-20 and ERC-721 token standards and participated in smart-contract security testing and reviews.

Blockchain Developer
Estatos

Jan 2025 - Feb 2025
Remote

- Developed and optimized smart contracts for blockchain operations.
- Worked on tokenization and DeFi protocols, contributing to secure and efficient smart-contract deployments.
- Collaborated with the tech team to enhance solutions while demonstrating problem-solving and a growth mindset.

Solidity Developer
Nexus Network

Jul 2024 - Dec 2024
Remote

- Developed bridge contracts for relayer-based cross-chain transactions.
- Strengthened smart-contract security and optimization practices.
- Actively participated in team collaborations and implemented feedback for continuous improvement.

EDUCATION

Post Graduate Diploma in Blockchain
IIT Bangalore & upGrad

Mar 2023 - Mar 2024
Bengaluru, IN

- Completed advanced coursework covering JavaScript, Node.js, Ethereum smart-contract development, and Hyperledger Fabric.
- Developed Ethereum-based smart-contract architectures for decentralized applications including DeFi systems.
- Designed system architecture for the UPSTAC healthcare application, focusing on scalable and secure blockchain integration.
- Explored blockchain-based KYC systems, Hyperledger Fabric network setup, and cross-chain frameworks.

Dual Degree (B.Tech + M.Tech), Energy Science & Engineering
Indian Institute of Technology Bombay (IIT Bombay)

Jul 2014 - Jul 2019
Mumbai, IN

- Secured 61%.

ADDITIONAL INFORMATION

- **Languages:** Fluent in Hindi (Native) and English (Professional)
- **Interests:** Reading and Cycling